

What is **HTML**?

↓
Hypertext markup language
↓
structure & formatting

HTML elements

↓
standard elements that
browser recognizes:

1. Paragraph <p> </p>
2. Heading <h1> </h1>
3. Image

▶ The component used to design
the structure of website are
called HTML tags

▶ HTML Tags

↓
Container for some content
or other tags

<p> This is a paragraph </p>



Eg:

▶ **Nesting**: when one tag
is used inside another
tag. That tag is called
as "Nested tags"

for eg: <p> This is my
 car </p>

▶ Heading Element

↓
<h1> to <h6> six level
headings

▶ HTML boilerplate

This is the standard format or
skeleton of writing HTML code

<!DOCTYPE html> (HTML5)

<html>

<head> ← meta data

<title> My first page </title>

</head>

<body>

<p> hello world </p>

</body>

</html>

▶ default code / boilerplate is
generated via **"emmet"**

↓
automatically syntax
code!

▶ "meta information" → extra
information about data

▶ ~~tab~~ Indentation: spaces for
good presentation of codes

▶ **Lists** in HTML → unordered list
→ ordered list

 Bread

 Butter

 Jam

 bread

 Butter

 Jam

► **HTML Attributes**
Attributes are used to add more information to the tag.
`<html lang = "en">`
or
`<ol type = "A">`
attribute → value

► **Anchor Element**
used to add links to your page
`<a href = "https://google.com">`
Google `</a>` → hypertext reference

- Absolute link
↓
specific link for information
- Relative link
↓
relative content link

► **Image Element**
used to add image to your page
`<img src = "image.png" alt = "Random Image">`
source → alternative

► absolute image / url (online image)
`<img src = "online-url" alt = "online image">`

► **Br Tag**
used to add next line (line breaks) to your page.
`<br>` ⇒ syntax
`<p>` Twinkle Twinkle little star `<br>` how I wonder what you are `</p>`

- Bold = `<b>` Bold `</b>`
- Italic = `<i>` italic `</i>`
- Underline = `<u>` underline `</u>`

► **Comment**
syntax: `<!-- comment -->`
ignored text by compiler/browser
shortcut: ctrl + / (forward slash)

Resources: MDN

► HTML is not case sensitive
`<html>` = `<HTML>`
`<p>` = `<P>`
`<head>` = `<HEAD>`

► **Block Elements**

↓
takes up the full width available (whole block)

start from new line

For eg: `<h1>` `</h1>`
`<p>` ... `</p>`

► **Inline Elements**

↓
takes up only necessary width

don't start from new line

For example:
Anchor tag,
Image element

div Element → block element
 → content division element
 is a container
 used to hold other HTML elements or group elements together.

<div>

 Google

 Youtube

</div>

 Facebook

O/P: Google Youtube
Facebook

► **Span Element** → inline element
 is also a generic container that is used to hold other HTML elements or group elements together

► **HR tag**
 Horizontal Rule Element
 <hr>

► **sub & sup Tag**
 subscript → superscript
 $a^2 = b^2 + c^2$ → super
 ²

H_2O → subscript H ₂ </sub> O

Eg:

<h1> Pythagoras theorem
 says $a^{²} + b^{²} = c^{²}$
 </h1>

► Semantic Markup

It is the markup that relates to the meaning of content

↙ semantic tags
 ↘ non-semantic tags

- <h1>
- <p>
- , <video>

- <div>
-

► Why Semantic tags are required?

- meaningfulness, layout structured

- seo friendly

↓
 search engine optimization

- readable (especially screen readers)

for eg: <header></header>

<main></main>

<footer></footer>

<nav></nav>

<article></article>

<section> </section>
<aside> </aside>
↓
related
content

► HTML entities

✓
An space / HTML entity is a piece of text ("string") that begins with an ampersand (&) and ends with a semi-colon (;)

► used to display reserved characters (which would otherwise be interpreted as HTML code), and invisible characters (like non-breaking spaces)

► can also use in place of characters that are difficult to type with a standard keyboard.

► browser interprets them & renders correct character

For eg : for & → &

For space →

↓
non breaking space

For < → <

> → >

For " → "

► NOTE : emmet - is
parent child : >
sibling : +
multiply : *

► Understanding HTML5

↓
set of modern web technologies which includes the HTML living standard & along with JS APIs to enhance storage, multimedia and hardware access

► How HTML works?

✓ HTML standard
is a document that tells to the browser how HTML should work.

► Video tag

<video controls width
="300px" height="300px"
source src="url" type=
"video/webm"/>
</video>

► Tables in HTML

↓
used to represent real life table data.

• Row & columns

Syntax:

```

<table>
  <caption> Table title </caption>
  <tr>
    <th> header 1 </th>
    <th> header 2 </th>
  </tr>
  <tr>
    <td> data 1 </td>
    <td> data 2 </td>
  </tr>
</table>

```

Annotations:
 - **table header** points to the `<tr>` containing `<th>` tags.
 - **table body** points to the `<tr>` containing `<td>` tags.
 - **table data** points to the `<td>` tags.

Semantics in tables

- `<thead>` to wrap table header
- `<tbody>` to wrap table body
- `<tfoot>` to wrap table footer

colspan & Rowspan

used to create cells which spans over multiple rows & columns.

`rowspan = " "`
`colspan = " "`

Forms in HTML

used to collect data from the user

Syntax:

```

<form>
  form content
</form>

```

Action Attribute

is used to define what action needs to be performed when a form is submitted or where form data should be sent

`<form action = "/action.php">`

Form Elements → **INPUT**

used to create multiple form controls

there are multiple types of inputs that can be used creating using type attributes

Syntax:

`<input type = "text">`

`<input type = "password">`

`<input type = "number">`

`<input type = "time">`

Place holders & Labels

`<input type = "text"`
`placeholder = "Enter Name">`

`<label>`

User name

`<input type = "text"`
`placeholder = "Anshika" />`

`</label>`

▶ Button Element

```
<button type="submit">
```

```
submit </button>
```

```
<button type="button"> submit
```

```
</button>
```

```
<button type="reset"> resetsubmit
```

```
</button>
```

▶ Name Attributes

```
<input type="text"
```

```
placeholder="Enter name"
```

```
id="username"
```

```
name="username" />
```

▶ Input Element - Checkbox

```
<input type="checkbox"
```

```
name="age" id="age"
```

```
checked /> by default display checked
```

```
<label for="age"> I am 18+
```

```
</label>
```

▶ Radio Element

```
<input type="radio"
```

```
name="fruit" id="apple"
```

```
value="apple" />
```

```
<label for="apple"> Apple </label>
```